
Software Design Document (SDD)

Project Title: Online Voting System for SRC Elections

Team Name: Online Voting System Development Team

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1. INTRODUCTION

1.1 Purpose

This Software Design Document (SDD) describes the architecture and system design of the Online Voting System for SRC Elections at the University of Eswatini. The purpose of the system is to allow students to vote electronically for Student Representative Council (SRC) candidates in a secure, transparent, reliable and efficient manner. The document follows the IEEE Std 1016-2009 guidelines.

1.2 Scope

The system is a secure web-based platform for conducting Student Representative Council (SRC) elections at the University of Eswatini. The system allows eligible students to: Log into the system using student credentials, View election information, View candidates, Cast votes electronically. The system also allows administrators to: Manage students, manage candidates, create elections, monitor voting progress, Generate election results. *Objectives*: Eliminate paper-based voting, reduce election fraud, improve accuracy of vote counting, increase accessibility and convenience, Provide faster election results. *Benefits*: Time saving, Reduced operational costs, Improved transparency, better election management, Increased student participation

1.3 Overview

This document explains the complete software design of the Online Voting System. It includes:

- System overview.
- System architecture
- Database design
- Components of the system
- Covers data design.
- User interface design
- Functional mapping of requirements
- Contains appendices.

1.4 Reference Material

The following materials were used:

- IEEE Std 1016 Software Design Description Standard
- Database Systems textbooks
- Web Application Development documentation
- SRS Document for Uneswa SRC Online Voting System
- OWASP Secure Coding Practices
- University of Eswatini Student Union Constitution (Election Guidelines)
- IEEE Std 830-1998: Software Requirements Specification

1.5 Definitions and Acronyms

Term/Acronym	Meaning
SDD	Software Design Document
SRC	Student Representative Council
UI/UX	User Interface / User Experience
DBMS	Database Management System
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
SQL	Structured Query Language
Admin	Election official or System Administrator
Authentication	Verification of user identity
Voter ID	Unique student identification number
Ballot	Set of choices for all positions in an election
Audit Log	Immutable record of all system actions
OTP	One-Time Password

2. SYSTEM OVERVIEW

The Online Voting System is a secure web-based application designed for conducting SRC elections online. The system operates in a client-server architecture with strong emphasis on security, scalability during peak voting hours, and compliance with university policies. Students access the system through a browser using their student credentials. Once authenticated, students can:

- View active elections
- See candidate profiles
- Vote for preferred candidates

The system ensures that:

- Each student votes only once
- supports multiple positions (President, Secretary, Faculty Representatives)
- Votes remain confidential
- Election results are accurately calculated

The administrator controls the election process by:

- Registering candidates
- Managing voters
- Starting and ending elections
- Viewing reports and results

The system uses:

- Frontend interface for users
- Backend server for processing
- Database for data storage

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

The system follows a Three-Tier Architecture:

1. Presentation Layer-Handles user interaction.

Components:

- Login page
- Voting dashboard
- Candidate pages
- Admin dashboard

2. Application Layer- Handles business logic.

Functions:

- Authentication
- Vote validation
- Result computation
- Election management

3. Data Layer- Stores and manages data. Database tables include:

- Students
- Candidates
- Elections
- Votes
- Administrators

3.2 Decomposition Description

Main Modules

1. Authentication Module

Responsibilities:

- Login validation
- Session management
- Access control

2. Student Module

Responsibilities:

- View candidates
- Cast vote
- Confirm vote

3. Election Management Module

Responsibilities:

- Create elections
- Open/close voting
- Manage election dates

4. Candidate Management Module

Responsibilities:

- Add candidates
- Edit candidate information
- Remove candidates

5. Result Management Module

Responsibilities:

- Count votes
- Generate results
- Display winners

6. Database Module

Responsibilities:

- Store system data
- Retrieve information
- Maintain data integrity

3.3 Design Rationale

The three-tier architecture was selected because:

- It improves scalability
- It separates system responsibilities
- It simplifies maintenance
- It enhances security

Alternative Considered

A two-tier architecture was considered but rejected because:

It provides less scalability

It mixes logic with presentation

It is harder to maintain

4. DATA DESIGN

4.1 Data Description

The system transforms election domain information into relational structures. Major entities are stored in a centralized secure database with encryption for sensitive fields. Votes are stored encrypted to preserve secrecy.

Main Data Entities

- Student
- Candidate

- Election
- Vote
- Administrator

Relationships

- One student casts one vote per election
- One election contains many candidates
- One candidate receives many votes

4.2 Data Dictionary

STUDENT Table

Field Name	Type	Description
student_id	Integer	Unique student identifier
full_name	Varchar	Student full name
email	Varchar	Student email
password	Varchar	Encrypted password
faculty	Varchar	Student faculty

CANDIDATE Table

Field Name	Type	Description
candidate_id	Integer	Unique Candidate identifier
full_name	Varchar	Candidate name
position	Varchar	Position contested
manifesto	Text	Candidate manifesto

ELECTION Table

Field Name	Type	Description
election_id	Integer	Election identifier
title	Varchar	Election title
start_date	Date	Election start date
end_date	Date	Election end date
status	Varchar	Active/Closed

VOTE Table

Field Name	Type	Description
vote_id	Integer	Vote identifier
student_id	Integer	Student voter
candidate_id	Integer	Selected candidate
election_id	Integer	Election participated
vote_time	Timestamp	Time vote was cast

ADMIN Table

Field Name	Type	Description
admin_id	Integer	Admin identifier
username	Varchar	Admin username
password	Varchar	Admin password

5. COMPONENT DESIGN

Authentication Module Pseudocode

Plain text

BEGIN

 INPUT username, password

 IF username and password are correct THEN

 Grant access

 ELSE

 Display error message

 ENDIF

END

Candidate management Module Pseudocode

Plain text

BEGIN

 INPUT candidate_details

 IF action = "Add" THEN

 SAVE candidate

 ELSE IF action = "Edit" THEN

```
    UPDATE candidate_info
ELSE IF action = "Remove" THEN
    DELETE candidate
ENDIF
END
```

Voting Module Pseudocode

Plain text

```
BEGIN
    DISPLAY candidates

    SELECT preferred candidate

    IF student has not voted THEN
        Save vote
        Display confirmation
    ELSE
        Display "Already Voted"
    ENDIF
END
```

Result Processing Module Pseudocode

Plain text

```
BEGIN
    COUNT votes for each candidate

    SORT candidates by highest votes

    DISPLAY election results
END
```

6. HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The system interface is simple, user-friendly, and responsive.

Voter Portal

- Login
- Dashboard
- View candidates
- Vote
- Logout

Administrator Portal

- Manage elections
- Manage candidates

- Mmonitor live voting
- View reports
- Publish results

Feedback messages include:

- Successful login
- Vote confirmation
- Error notifications

6.2 Screen Images

Login Screen

```

-----
| UNIVERSITY SRC VOTING SYSTEM |
-----
| Student ID: [ ] |
| Password : [ ] |
|               |
| [ LOGIN BUTTON ] |
-----

```

Voting Dashboard

Plain text

```

-----
|  ACTIVE ELECTION  |
-----
| Candidate 1 [Vote] |
| Candidate 2 [Vote] |
| Candidate 3 [Vote] |
-----

```

Admin Dashboard

Plain text

```

-----
| ADMIN CONTROL PANEL |
-----
| Add Candidates      |
| Manage Elections    |
| View Results        |
| Generate Reports     |
-----

```

6.3 Screen Objects and Actions

Object	Action
--------	--------

Login Button	Authenticates user
Vote Button	Submits vote
Logout Button	Ends session
Add Candidate	Saves candidate details
Publish Results	Displays election results

7. REQUIREMENTS MATRIX

Requirement ID	Requirement Description	Component
FR1	Student login	Authentication Module
FR2	View candidates	Student Module
FR3	Cast vote	Voting Module
FR4	Prevent double voting	Voting Module
FR5	Generate results	Result Module
FR6	Manage candidates	Candidate Module
FR7	Manage elections	Election Management Module

8. APPENDICES

Appendix A: Tools and Technologies

Frontend

- HTML
- CSS
- JavaScript

Backend

- PHP / Python / Node.js

Database

- MySQL

Development Tools

- Visual Studio Code
- XAMPP
- GitHub

Appendix B: Security Features

Password encryption

Session timeout

Authentication validation

Access control

Vote confidentiality

Appendix C: Future Improvements

Mobile application support

Biometric authentication

SMS/email notifications

Real-time analytics dashboard

Blockchain-based vote verification